

Climate-Aware Crop Yield Prediction System

Using Machine Learning and Satellite Data

This project presents a climate-aware crop yield prediction framework using machine learning, satellite-derived vegetation indices, and historical weather information. The system integrates NDVI satellite data, district-level weather patterns, and agricultural production data to predict crop yield across districts of Uttar Pradesh.

1. Project Objectives

Predict crop yield using machine learning techniques. Integrate climate, satellite, and agricultural datasets. Detect climate anomalies affecting agricultural productivity. Apply feature engineering for improved prediction performance. Use explainable AI techniques for model interpretation.

2. Datasets Used

Dataset	Description
full_up_weather.csv	District-wise weather dataset containing temperature, humidity, and rainfall.
FULL_UP_District_NDVI_2015_2024.csv	Satellite NDVI dataset extracted from Google Earth Engine.
full_up_yield.csv	District-wise crop yield dataset.
final_large_scale_dataset.csv	Merged master dataset used for model training.
feature_engineered_dataset.csv	Enhanced dataset with anomaly and temporal features.

3. System Methodology

Collected district-wise crop production data. Generated weather datasets using NASA POWER climate parameters. Extracted NDVI values using Google Earth Engine and Sentinel-2 imagery. Merged datasets into a large-scale climate-yield dataset. Performed feature engineering including anomaly detection and rolling averages. Trained multiple machine learning models. Applied SHAP explainability analysis. Implemented dynamic climate correction mechanism.

4. Engineered Features

Temperature Anomaly Rainfall Anomaly Humidity Anomaly Climate Anomaly Score Previous Year Yield Rolling Temperature Average Rolling Rainfall Average District Encoding

5. Machine Learning Models

The following machine learning models were trained and evaluated: Random Forest Regressor XGBoost Regressor LightGBM Regressor CatBoost Regressor

6. Experimental Results

Model	MAE	RMSE	R ² Score
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Random Forest	0.2845	0.3880	0.6611
XGBoost	0.3008	0.4076	0.6259

7. Explainable AI (SHAP Analysis)

SHAP (SHapley Additive exPlanations) was used to interpret feature importance. The analysis showed that temperature, NDVI, and rainfall were the most influential factors affecting crop yield prediction.

8. Climate Anomaly Detection

Climate anomaly detection was implemented using engineered climate deviation features. Districts were categorized into: Normal Moderate Risk High Risk based on anomaly severity.

9. Final Pipeline

Crop Production Data + Weather Data + Satellite NDVI Data → Large Scale Merge → Feature Engineering → Climate Anomaly Detection → Advanced ML Training → SHAP Explainability → Dynamic Climate Correction → Final Yield Prediction

10. Conclusion

The project successfully developed a climate-aware crop yield prediction system using machine learning, satellite data, and weather information. The framework demonstrated strong prediction capability and introduced explainable, climate-sensitive forecasting mechanisms suitable for agricultural analytics and smart farming systems.

11. Future Scope

Integration with real-time weather APIs. Deployment as a web-based prediction dashboard. Incorporation of deep learning and LSTM-based forecasting. Use of multi-crop and soil datasets. Expansion to all Indian states.